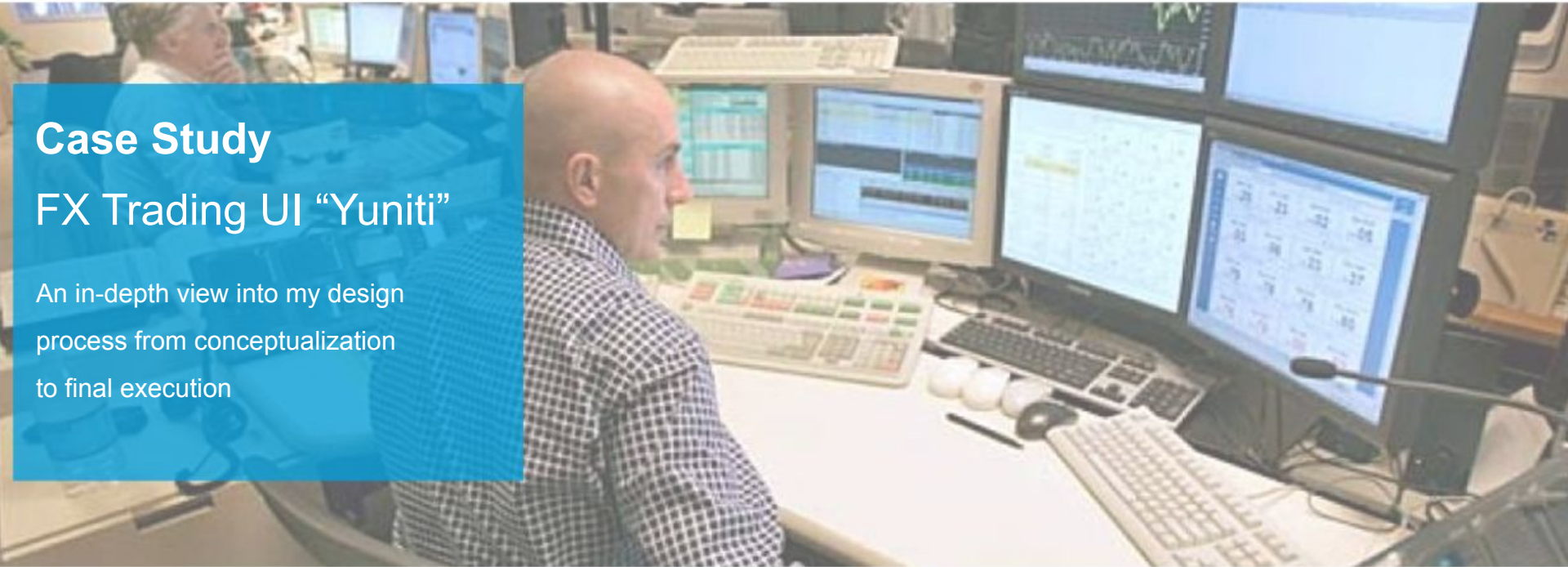


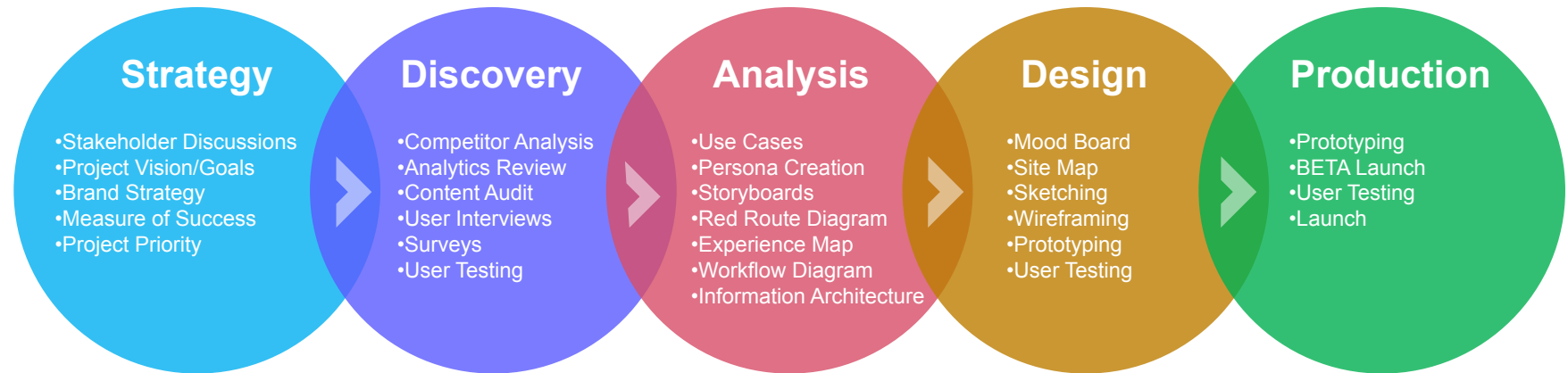
Case Study

FX Trading UI “Yuniti”

An in-depth view into my design process from conceptualization to final execution



Full UX design methodology



Included in this case study

- Stakeholder Discussions
- Project Vision/Goals
- Brand Strategy
- Measure of Success
- Project Priority

- Competitor Analysis
- Analytics Review
- Content Audit
- User Interviews
- Surveys
- User Testing

- Use Cases
- Persona Creation
- Workflow Diagram
- Information Architecture

- Site Map
- Sketching
- Wireframing
- Prototyping
- User Testing

- Prototyping
- BETA Launch
- User Testing
- Launch

Case Study

YUNITI TRADING UI

Project Introduction

- Industry:** Foreign Exchange trading transactional flow averages \$5.1 trillion per day.
- Company:** EBS BrokerTec averages \$100-\$200 billion per day, this project added \$10 billion per day in 6 months... and growing.
- Objective:** Provide customers with personalized user interface, integrated visual data, price depth and an ability to compete with algorithms.



Product Description:

Yuniti delivers exceptional customization for a user's individual trading needs.

Yuniti is a Foreign Exchange spot and forwards trading application designed specifically for professional traders at major investment banks, corporates and government agencies.

Yuniti's application is a streamlined browser based product delivered through an intuitive, responsive HTML5 user interface.

Equipped with advanced price discovery and efficient trade workflow, Yuniti provides access to market intelligence to inform strategy and manage risk.

Project Summary

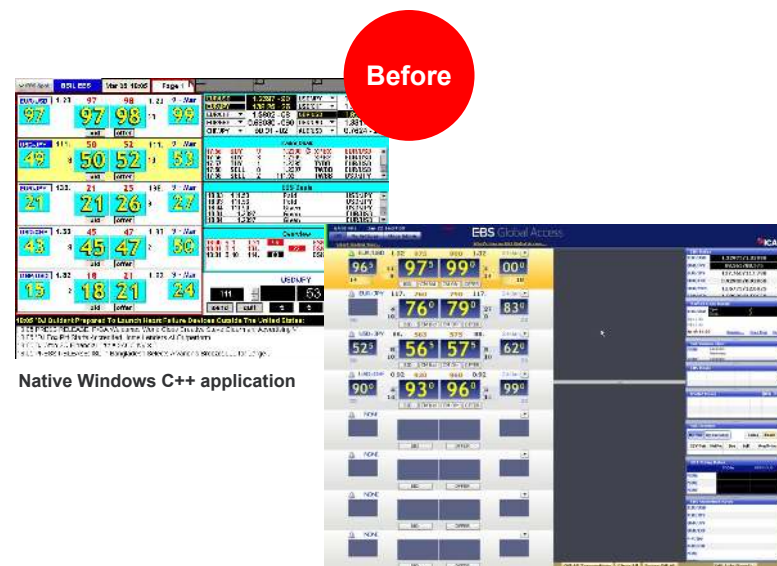
As a key collaborator my role as UX Designer was to research, design and test iteratively throughout the product life cycle.

From concept to execution I worked closely with users, stakeholders, product managers, 3rd party agencies, developers and QA to design this product.

After analyzing user research, I created sketches and low-fidelity wireframes with the Chief Technology Officer.

Afterward, I created high-fidelity wireframes in Photoshop to show default and interactive states. Working with the product manager, we used the wireframes to discuss product strategy and inserted the screens in JAMA, our SaaS business requirements management tool.

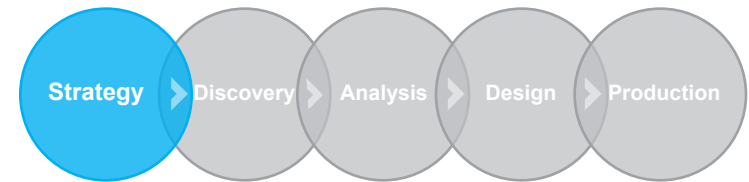
Before and after product screens:



HTML5 based application

Adobe Flex based application

Strategy



Stakeholder Discussions

Multiple interviews via conference calls and F2F meetings with key stakeholders.

Contributors:

- Executive management
- Customers
- Business-line owners
- Sales
- Marketing
- Technology
- Operations

Project Vision/Goals

Expand our market reach into FX Outrights, Swaps and Options thereby increasing our overall trading volume and fee revenue to grow our market share and bottom-line.

Through a new, attractive, intuitive user interface, we will provide customers with personalized trading, integrated visual data, price depth and an ability to compete with algorithms.

Brand Strategy

Create a state-of-the-art UI while still preserving elements of our product heritage.

Adhere to the latest brand standards.

Insure product appeal to our current customers as well as the next generation of traders.

Measure of Success

Success metrics:

- High product adoption rate.
- Inflow of new customers.
- Increase in trades and average daily volume.

Project Priority

Top priority for all teams globally.

Discovery > Competitor Analysis



Perform competitive research by creating profiles containing functions and features.

- Garnered client screenshots to understand how they utilized multiple monitors, products and screens.
- Evaluated competing product manuals and screen designs.
- Interviewed traders on product pros and cons and competitor site comparisons.

Profiles of some of our top competitors:

Thompson Reuters – Eikon

The screenshot displays the Bloomberg terminal interface with several key components:

- Top Section:** A navigation bar with icons for Home, News, Data, and other functions.
- Main Display Area:**
 - Left Panel:** A list of market indices and their current values, including S&P 500, NASDAQ, and others.
 - Center Panel:** A line chart showing the performance of a specific index or stock over time, with a green line and a red shaded area.
 - Right Panel:** A table of financial data, including columns for Name, Price, Change, and Volume.
- Bottom Section:** A status bar showing the user's name, session time, and other system information.

GTF – Dealbook 360

Product Highlights:

- Tracks assets with over 60 technical indicators, night chart types and a wide range of indicators
- Organize your trading space with custom layouts, so the position scheduler suits how you like it
- Real-time news from industry leading sources, including Reuters® and Bloomberg Global Markets
- Real-time analysis including Morning Star® insight, which is free to GTF customers
- Provides you with easy-to-read charts, a bundle of live indicators and a wide selection of customization options to help you do it all

View

Includes Quote board, Open positions, Working Orders and Charts

- Trade View – simple charts
- Trend View – charts with moving averages
- Dynamic Trend View – charts with Bollinger bands
- Pivot View – charts with pivots
- Pivot View – charts with support and resistance
- Break View – sources include Industry Global Markets and Reuters

Coverage

- FX – Confirms amount of spot, futures
- Index – multiple countries
- Commodity – Oil, Metals, Soft, Energy
- Bond – multiple countries
- Interest Rates

Web:

<http://www.gtf.com/Products/Dealbook360.aspx>

Video Demo:

<http://www.gtf.com/Products/Dealbook360.aspx#demo>

Discovery > Competitor Analysis

(continued)



- We found traders were stitching together their own system with numerous screens from different platforms.
- A tailored desk setup reinforced traders' virtuoso identity. Objects were physically arranged to map traders' muscle memory thereby minimizing risk for errors and maximizing economy of time.
- The standalone EBS workstation represents stability in it's separation.

The monitor setups of our customers:



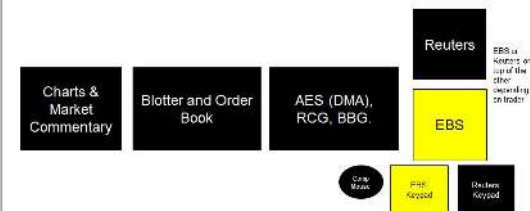
Institution: ABCL / Trader: Perry



Institution: Natixis / Trader: Anonymous



Institution: Credit Suisse / Trader: Anonymous

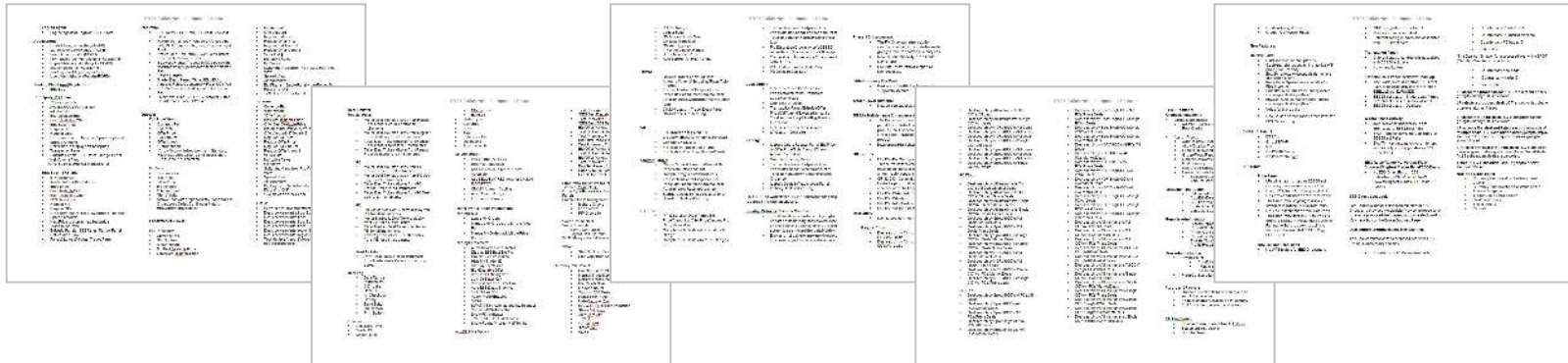


Discovery > Content Audit



Created a comprehensive list of function and features with the Quality Assurance team.
This included functions and features required for the new trading platform and the administrative system.

Snapshots of the function/feature set:



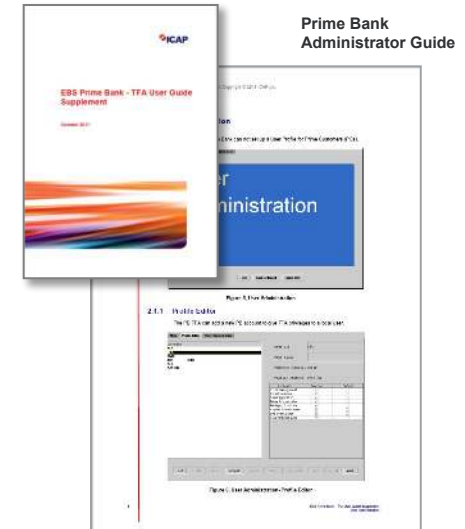
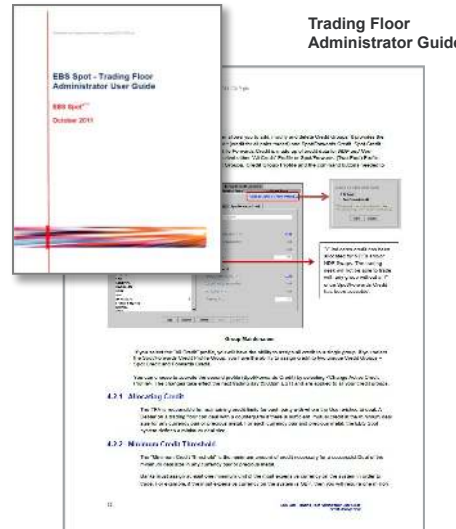
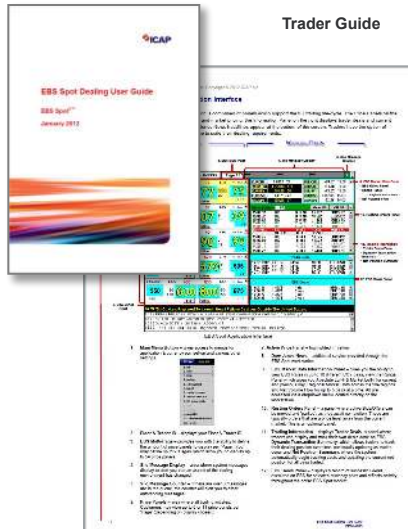
Discovery > Content Audit

(continued)



Studied the user manual written by the System Architecture team to research core trading workflow as well as system nuances.

Snapshots of our product manuals:



Discovery > User Interviews

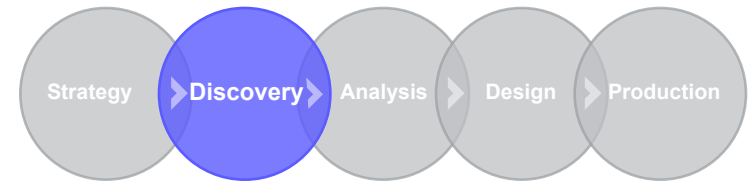


To get a deeper understanding of the user perspective, we utilized a 3rd party firm to facilitate user interviews giving us a clear, unfiltered, unbiased response from our customers. **The interviews were conducted at 44 banks with 62 traders in New York, London, Tokyo, Hong Kong and Singapore.** Collected data from these interviews consisted of **750+ pages of transcripts** and **30 hours of recorded interviews** which were then distilled to salient points in a final report.

Snapshots of the Global Research Report:



Discovery > Surveys



Working with marketing, sales and customer support teams, we created online surveys to gain an understanding of the limitations, gaps and usefulness of the current product.

We compiled the surveys in a final report that outlined key takeaways, comments and action items for our first deliverable.

Snapshots of the survey, responses and conclusion:

[illegible][illegible]

EBS

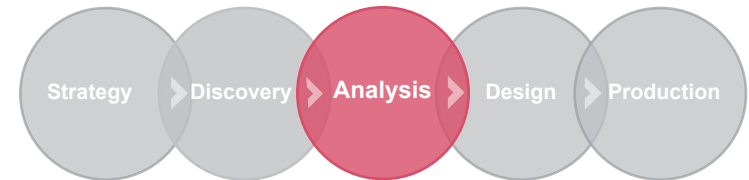


Conclusion:

- Health care services are vital
- Financial and social support are essential for the success of health care services
- Financial support is essential for the success of health care services
- Social support is essential for the success of health care services
- Quality of care is essential for the success of health care services
- Health care services are essential for the success of health care services
- Health care services are essential for the success of health care services
- Health care services are essential for the success of health care services

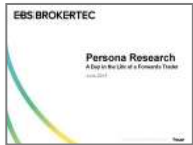
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Analysis > Persona Creation



Created target customer personas to **capture demographics, personalities, daily objectives** and **understand customers behavior** through a forwards trading scenario.

Snapshots of the persona document:



EBS BROKERTEC

Meet the Forwards Trader



Fictional Name: Frank
Job title/major responsibilities: Forwards trader at Big Bank X in London with 20 years of experience of trading "Forex"
Demographics:

- Male
- College educated
- Married
- Father

Goals and tasks:

- Make money for his customers trading forwards, firm and to hedge risk
- As a forwards trader, Frank trades a currency futures contract that is a legally binding contract that obligates the two parties involved to trade a particular amount of a currency pair at a predetermined price (the stated exchange rate) at some point in the future.

Environment:

- He uses internal and external systems for technical trading patterns, global financial news and chat features to communicate with his customers and peers. Some of which include Bloomberg's terminal, Reuters, EBS and a few single bank platforms.
- He accesses these systems over the internet and private lease lines.
- He is constantly scanning these systems for major trends in the foreign exchange market

EBS BROKERTEC

Meet the Spot Trader



Fictional Name: Bob
Job title/major responsibilities: Spot trader at Big Bank Y in London with 20 years of experience of trading "Forex"
Demographics:

- Male
- College educated
- Married
- Father

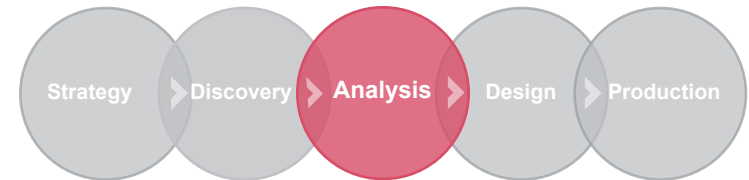
Goals and tasks:

- As spot trader, the underlying currencies are physically exchanged following the settlement date. In general, any spot market involves the actual exchange of the underlying asset.

Environment:

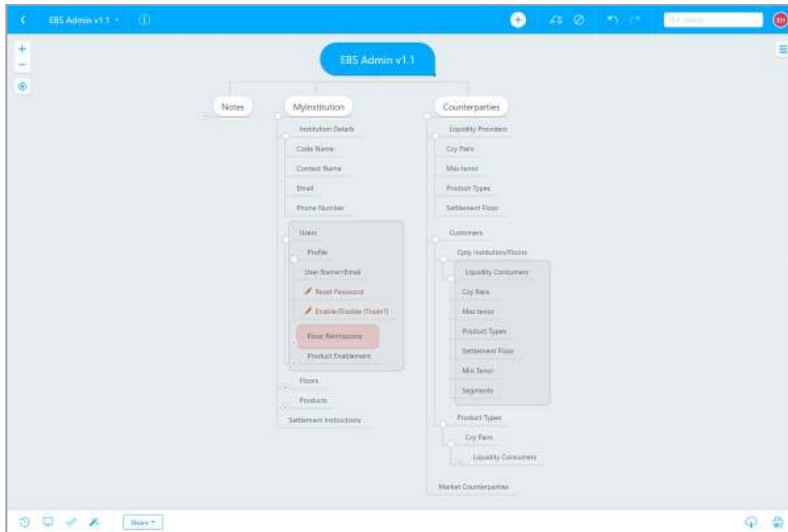
- He uses internal and external systems for technical trading patterns, global financial news and chat features to communicate with his customers and peers. Some of which include Bloomberg's terminal, Reuters, EBS and a few single bank platforms.
- He accesses these systems over the internet and private lease lines.
- He is constantly scanning these systems for major trends in the foreign exchange market and makes trading decisions accordingly.

Analysis > Information Architecture

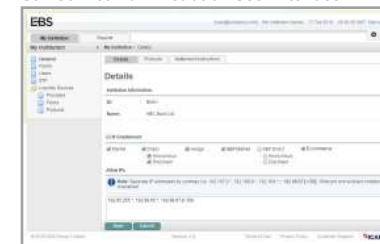


Working with a diverse cross-functional group, I **created an information architecture diagram using a collaborative SaaS application called “Mindmeister”**. The diagram visualized the customer self-service application that would be used to configure the trading UI parameters for the institution on a per group and per user basis.

Interactive IA diagram trading and administrative ecosystem:



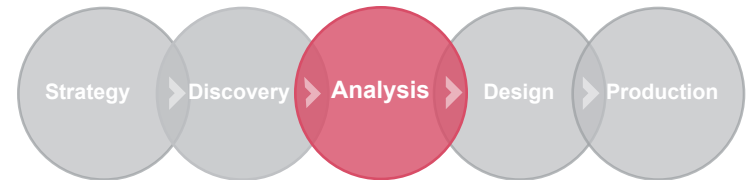
Self-Service Administration User Interface



Trading User Interface



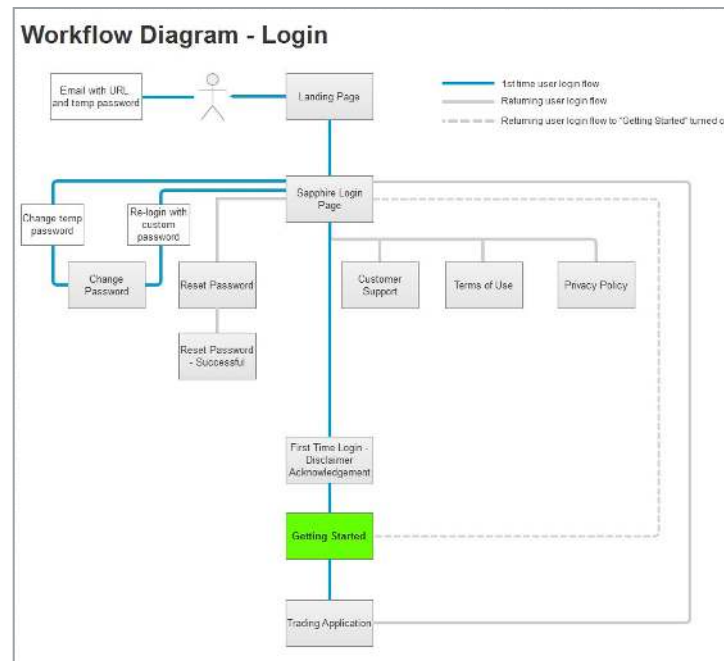
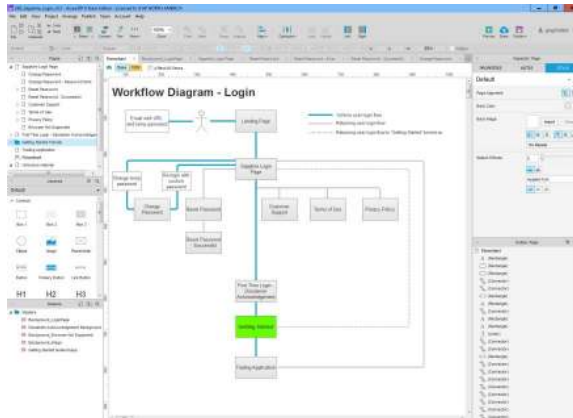
Analysis > Workflow



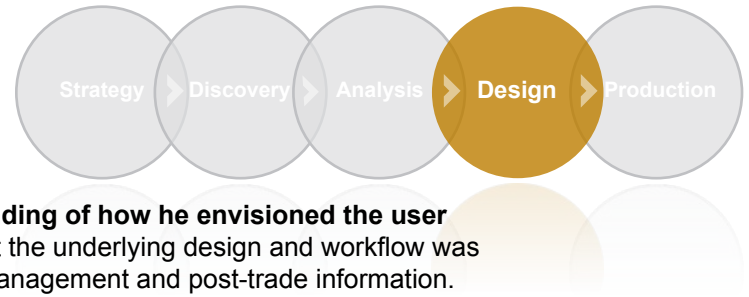
Using Axure RP, I created workflow diagrams to visualize first-time and existing user login experiences.

The experience had to capture requirements from legal, marketing, compliance and product management such as email verification, entry point, password reset, customer support links, user acknowledgement for usage and a simple product tour for first-time users.

Customer login workflow diagram in Axure:

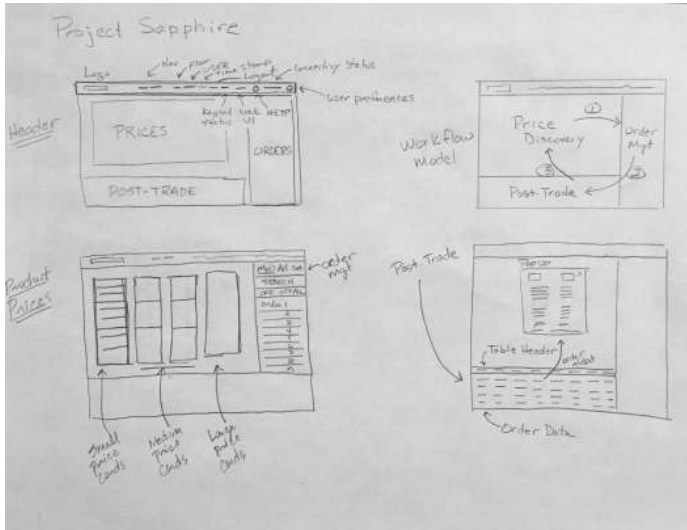


Design > Sketches



Collaborated with the CTO to create sketches in order to get a better understanding of how he envisioned the user interface and experience. Although the content was complex, I wanted to show that the underlying design and workflow was extremely easy. Traders had 4 main areas of focus: Header, price discovery, order management and post-trade information.

Sketches of the basic UI anatomy and trading workflow:

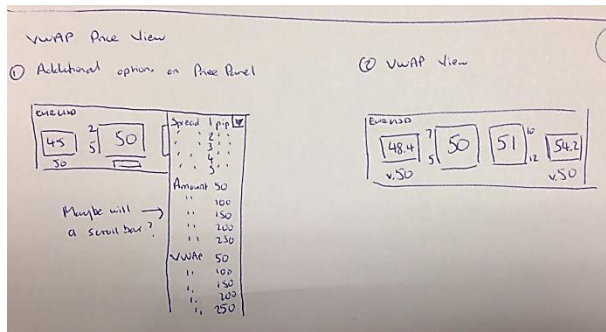


Design > Sketches

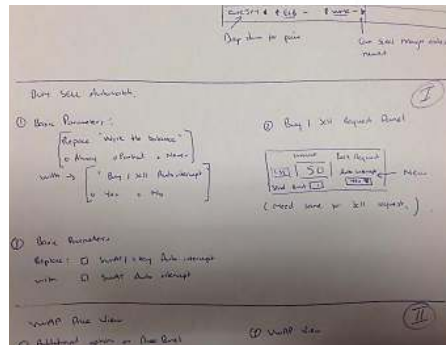
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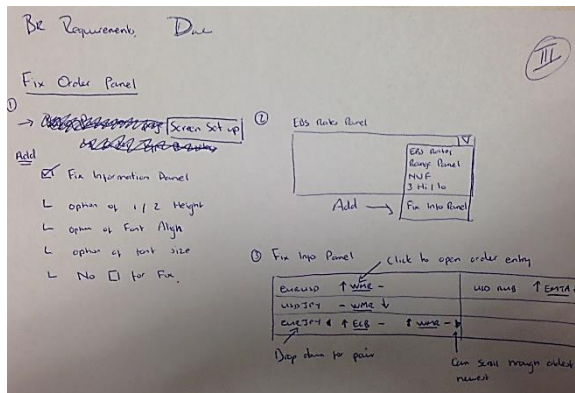
Early price panel sketches



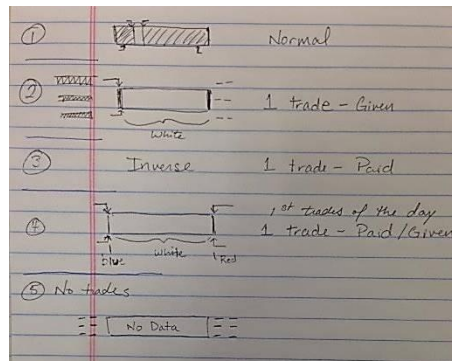
Early price panel sketches



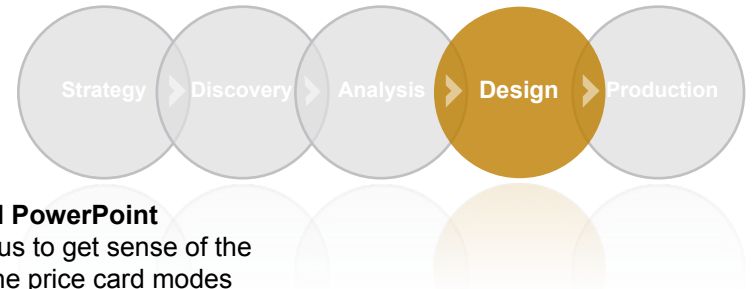
Early price panel sketches of fixing order menus



Early price panel sketches of the market activity data visualizations



Design > Lo-Fi Wireframing



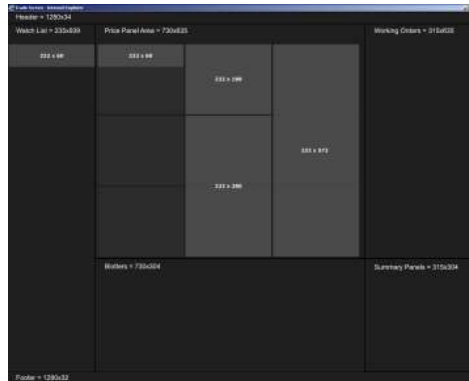
For the low fidelity wireframe phase, I utilized Adobe Illustrator, Photoshop and PowerPoint to create the general layout of the application. The low fidelity mock-ups allowed us to get sense of the basic display model, general layout, optimal screen resolution and basic workflow. The price card modes were designed in PowerPoint so I could collaborate with the CTO who was familiar with this tool.

Low fidelity wireframe artifacts:

General layout wireframe



General layout Lo-Fi wireframe with pixel dimensions



Lo-Fi wireframes of medium price panel cards



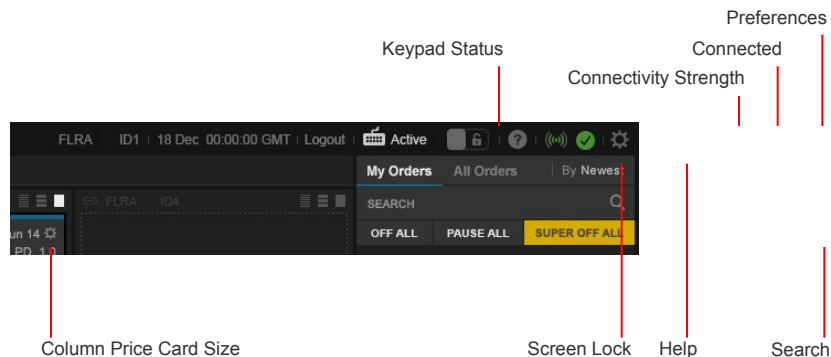
Design > Iconography



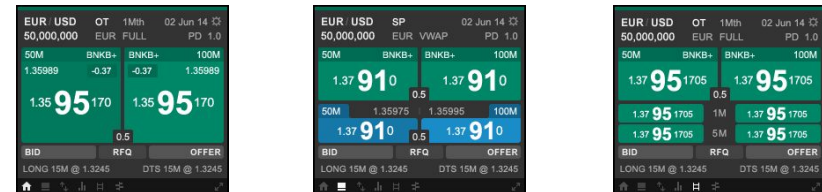
Although some of the icons in this user interface are universal, I was required me to create unique icons that were specific to this project. The price panel icon (bottom right) had abstract concepts that needed to be designed to symbolized various additional views that supplemented the main price (or top of book price). In preliminary user testing, the icons were not obvious. However, once the users clicked on the icons in the interactive prototype, **their meaning became apparent and did not have to be relearned.**

Iconography examples:

Header & Column Icons



Price Card Icons



Design > Hi-Fi Wireframing



In the high fidelity phase, I utilized Adobe Illustrator & Photoshop to iterate several times until we had a final version. The high fidelity wireframes gave visual clarity to navigation, type styles, font sizes, color usage and overall design treatment.

High fidelity wireframe examples:

Hi-Fi Wireframes – 1st iteration



Hi-Fi Wireframes – 19th iteration

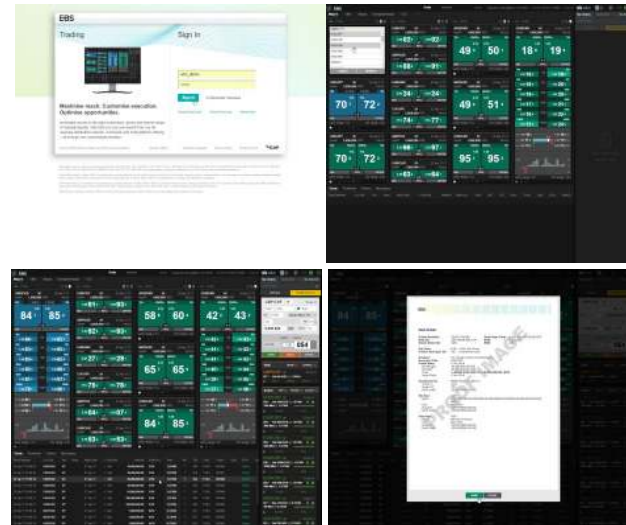
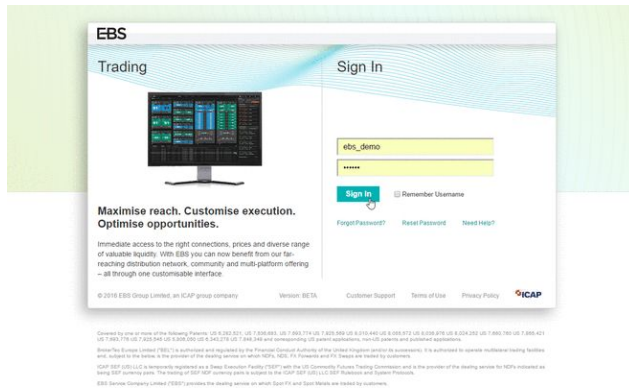


Design > Interactive Prototype



To code the interactive prototypes I hired and trained a UI developer. The UI developer used the following technology stack: Angular, Grunt, Node, Ruby and Git. The interactive prototype helped articulate interaction, button states and system feedback messages.

An animated GIF file accompanied by screen shots of the user interaction:



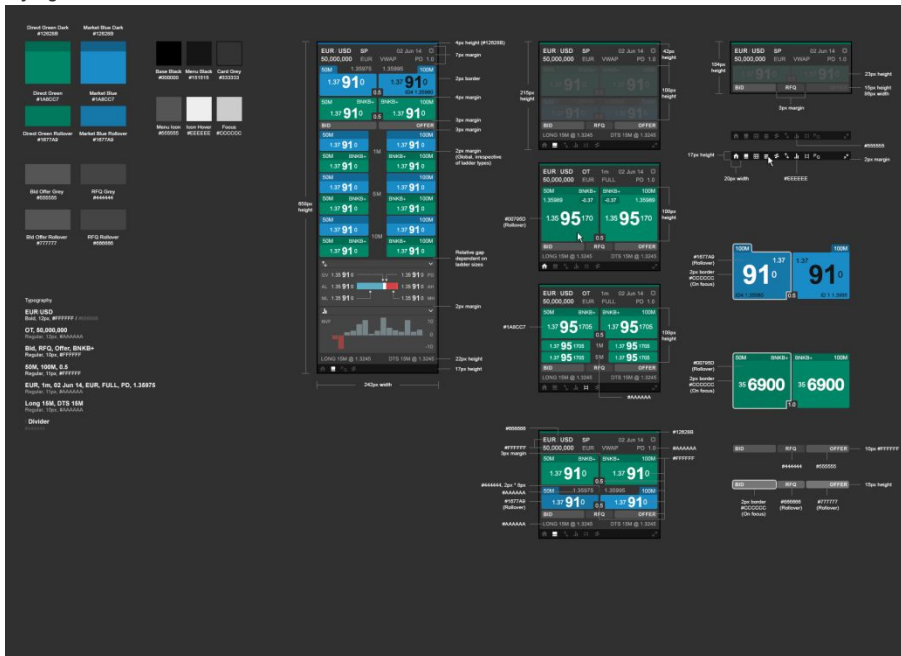
Production > Style Guide



Utilizing Photoshop and Illustrator I created color palettes and design specifications for each UI component.

Style guide examples:

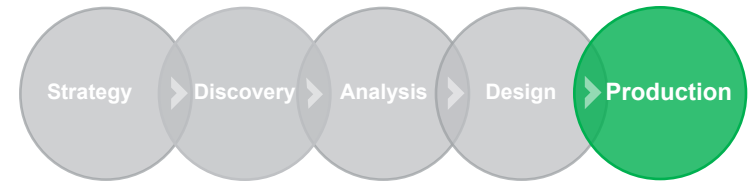
Style guide for Price Cards



Style guide for user preference forms



Production



Development Prototype

Prototype development proceeded with our on-shore team and offshore team in Tel Aviv. We performed on-site and off-site spot checks for quality control.

We performed stress tests every 2 weeks in our "Global Test Flights". We wrote test scripts for internal users to perform activities simultaneously worldwide.

Defects and errors were recorded in JIRA. The development team corrected the defects and another global test flight would follow.

BETA Launch

We initiated our BETA launch in 2016 with several customers in our pilot program.

User Testing

We used the System Usability Scale (SUS) questionnaire to quantify satisfaction. We found a high satisfaction rate of approximately 90%. Approximately 20 users from the pilot launch responded.

In addition, we received feedback from the pilot group on a daily basis. Our account managers logged customer comments in Sale Force reports. Our team analyzed the commentary and prioritized them for future versions.

Launch

The company had a soft launch in Q4 of 2016 and a full launch in Q1 of 2017.

To date, the launch has been a success and currently contributes \$10 billion in transactions daily which makes up 10% of the average daily trading volume.

Post-Launch

Post-launch activity included:

- Creating a video demo
- Internal training micro-site
- Online product tour
- Close monitoring of customer feedback in Salesforce.com

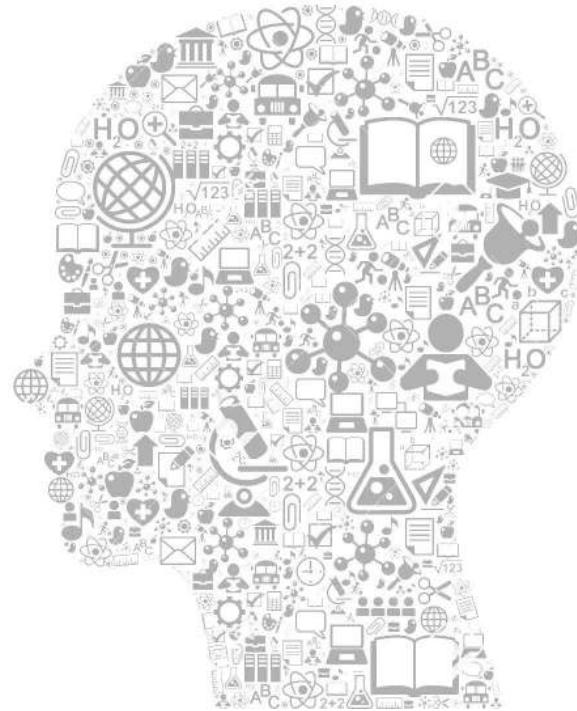
The full breadth and depth of our legacy offering is now being integrated which will take at least 5 years.

What did I learn?

I took great pride in owning the design for the Yuniti Trading platform, playing an essential role in its daily development. While the workload was heavy, our team successfully managed priorities to meet our goals.

Despite initial customer resistance to replacing our inefficient legacy system, they quickly embraced the new, intuitive features—specifically price depth of view and improved customizability. Through targeted research, we refined our design to better serve existing users and our new segment, Forwards Traders.

The platform's success is measured by strong adoption: within six months of launch, our average daily volume grew from \$0 to \$10 billion, and we onboarded numerous institutions, including many legacy clients. Creating this innovative, highly acclaimed product remains a defining achievement in my career.



Contact & Recommendations



Rick Hollandsworth
Senior UX Designer

Contact:

(201) 404-5031

rick.hollandsworth@outlook.com

<https://www.linkedin.com/in/rickhollandsworth/>



Gary Brunkhorst

Director, Transaction Sales -
Thomson Reuters

March 27, 2017, Rick worked with
Gary in the same group

Rick carries with him unsurpassed expertise in the areas of user design and user experience. As part of the product team he was responsible for designing new screens for the EBS trading platform. He had the uncanny ability, and patience, to visual the team's business requirements and create an aesthetically intuitive interface. Rick is passionate about his work and eager to learn more about the business. Both are reflected in what he produces. It was a pleasure working with him.



Scott Leavitt

Sales and Marketing
Excellence at BASF

January 17, 2010, Scott managed
Rick directly

Rick reported into me on a portal strategy project for a large financial services firm. During this engagement, he was required to stretch beyond his normal site design role, to one in which he needed to develop and communicate a set of usability recommendations to a business audience. I was quite impressed by Rick's ability to grow into this broader consultative role, and his willingness to seek out, internalize and apply constructive feedback. This is indicative of Rick's strong desire to continuously learn and grow professionally.



Shilpa Patange CUA

Sr. User Experience Designer
at HID Global

April 1, 2008, Shilpa worked with
Rick in different groups

Rick and I work in partnership to cater to business with UI needs. He is a very talented individual and a very good team-mate. He has set the bar high for perfection and always shoots for the best UI solution. He uses his previous experience in UI domain on a perfect situation. He has an abundant innovative ideas to handle various situations. He is been very helpful and positive. We have been a successful partnership in terms of work distribution, problem solving, and discussing design solutions and taking decisions on UI standards. Besides, I do want to mention that he is very good human being, kind and warm hearted.